

Susceptibility and interactions between Aedes mosquitoes and Zika viruses.

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Zika virus disease is caused by Zika virus infection, as transmitted by Aedes spp. mosquitoes. Many of the Zika virus strains isolated from patients display different pathogenicities toward humans. The vector mosquitoes for Zika virus are mainly of the Aedes genus, especially Aedes aegypti and Aedes albopictus. However, susceptibility and interactions between Aedes spp. mosquitoes and Zika viruses remain unclear. In this study, we chose two Zika virus strains (FSS13025 and PRVABC59) with different abilities to infect the primary vector mosquitoes Ae. aegypti and Ae. albopictus. The transcriptomes and small RNA profiles of infected and uninfected mosquitoes were comparatively analyzed, and differentially expressed genes were functionally examined using RNA interference. According to the results, the susceptibility of PRVABC59 was higher than that of FSS13025 in Aedes vector mosquitoes, and Ae. aegypti was more susceptible to Zika virus than was Ae. albopictus. For PRVABC59 infection, specific differential expression profiles correlated with Ae. aegypti and Ae. albopictus, and susceptibility was significantly affected when three targeted genes were successfully knocked down. Compared with PRVABC59, infection of Ae. albopictus with FSS13025 generated more 21-nt virus small interference RNA. It can be concluded that the susceptibility of vector Aedes spp. mosquitoes to Zika viruses varies and that the interactions between mosquitoes and Zika virus correlate with susceptibility.

American Aedes vexans Mosquitoes are Competent Vectors of Zika Virus.

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AbstractStarting in 2013-2014, the Americas have experienced a massive outbreak of Zika virus (ZIKV) which has now reached at least 49 countries. Although most cases have occurred in South America and the Caribbean, imported and autochthonous cases have occurred in the United States. Aedes aegypti and Aedes albopictus mosquitoes are known vectors of ZIKV. Little is known about the potential for temperate Aedes mosquitoes to transmit ZIKV. Aedes vexans has a worldwide distribution, is highly abundant in particular localities, aggressively bites humans, and is a competent vector of several arboviruses. However, it is not clear whether Ae. vexans mosquitoes are competent to transmit ZIKV. To determine the vector competence of Ae. vexans for ZIKV, wild-caught mosquitoes were exposed to an infectious bloodmeal containing a ZIKV strain isolated during the current outbreak. Approximately 80% of 148 mosquitoes tested became infected by ZIKV, and approximately 5% transmitted infectious virus after 14 days of extrinsic incubation. These results establish that Ae. vexans are competent ZIKV vectors. Their relative importance as vectors (i.e., their vectorial capacity) depends on feeding behavior, longevity, and other factors that are likely to vary in ecologically distinct environments.

Competence of Aedes aegypti, Ae. albopictus, and Culex quinquefasciatus Mosquitoes as Zika Virus Vectors, China.

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In China, the prevention and control of Zika virus disease has been a public health threat since the first imported case was reported in February 2016. To determine the vector competence of potential vector mosquito species, we experimentally infected Aedes aegypti, Ae. albopictus, and Culex quinquefasciatus

mosquitoes and determined infection rates, dissemination rates, and transmission rates. We found the highest vector competence for the imported Zika virus in *Ae. aegypti* mosquitoes, some susceptibility of *Ae. albopictus* mosquitoes, but no transmission ability for *Cx. quinquefasciatus* mosquitoes. Considering that, in China, *Ae. albopictus* mosquitoes are widely distributed but *Ae. aegypti* mosquito distribution is limited, *Ae. albopictus* mosquitoes are a potential primary vector for Zika virus and should be targeted in vector control strategies.

Effects of Zika Virus Strain and Aedes Mosquito Species on Vector Competence.

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In the Western Hemisphere, Zika virus is thought to be transmitted primarily by *Aedes aegypti* mosquitoes. To determine the extent to which *Ae. albopictus* mosquitoes from the United States are capable of transmitting Zika virus and the influence of virus dose, virus strain, and mosquito species on vector competence, we evaluated multiple doses of representative Zika virus strains in *Ae. aegypti* and *Ae. albopictus* mosquitoes. Virus preparation (fresh vs. frozen) significantly affected virus infectivity in mosquitoes. We calculated 50% infectious doses to be 6.1-7.5 log₁₀ PFU/mL; minimum infective dose was 4.2 log₁₀ PFU/mL. *Ae. albopictus* mosquitoes were more susceptible to infection than *Ae. aegypti* mosquitoes, but transmission efficiency was higher for *Ae. aegypti* mosquitoes, indicating a transmission barrier in *Ae. albopictus* mosquitoes. Results suggest that, although Zika virus transmission is relatively inefficient overall and dependent on virus strain and mosquito species, *Ae. albopictus* mosquitoes could become major vectors in the Americas.

Zika vector competence data reveals risks of outbreaks: the contribution of the European ZIKAlliance project.

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First identified in 1947, Zika virus took roughly 70 years to cause a pandemic unusually associated with virus-induced brain damage in newborns. Zika virus is transmitted by mosquitoes, mainly *Aedes aegypti*, and secondarily, *Aedes albopictus*, both colonizing a large strip encompassing tropical and temperate regions. As part of the international project ZIKAlliance initiated in 2016, 50 mosquito populations from six species collected in 12 countries were experimentally infected with different Zika viruses. Here, we show that *Ae. aegypti* is mainly responsible for Zika virus transmission having the highest susceptibility to viral infections. Other species play a secondary role in transmission while *Culex* mosquitoes are largely non-susceptible. Zika strain is expected to significantly modulate transmission efficiency with African strains being more likely to cause an outbreak. As the distribution of *Ae. aegypti* will doubtless expand with climate change and without new marketed vaccines, all the ingredients are in place to relive a new pandemic of Zika.

Vector competence and transovarial transmission of two *Aedes aegypti* strains to Zika virus.

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Zika virus (ZIKV) has become a serious threat to global health since the outbreak in Brazil in 2015. Additional Chinese cases have continuously been reported since the first case of laboratory-confirmed ZIKV infection in China on 6 February 2016. *Aedes aegypti* is the most important vector for ZIKV. This study shows that two strains from China exhibit high levels of midgut infection and highly disseminated infection of salivary glands and ovaries. Both strains can transmit ZIKV to infant mice bitten by infectious mosquitoes. Moreover, the results provide the evidence of transovarial transmission of ZIKV in mosquitoes. The study indicates that the two *Ae. aegypti* strains are not only effective transmission vectors but also persistent survival hosts for ZIKV during unfavorable inter-epidemic periods. This function as a reservoir of infection has epidemiological implications that further enhance the risk of potential future outbreaks.

Vector competence of selected North American *Anopheles* and *Culex* mosquitoes for Zika virus.

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Zika virus (ZIKV) is a vector-borne flavivirus that has caused recent outbreaks associated with serious disease in infants and newborns in the Americas. *Aedes* mosquitoes are the primary vectors for ZIKV, but little is known about the diversity of mosquitoes that can transmit ZIKV in North America. We chose three abundant North American mosquito species (*Anopheles freeborni*, *Anopheles quadrimaculatus*, and *Culex tarsalis*) and one known vector species (*Aedes aegypti*), fed them blood meals supplemented with a recent outbreak ZIKV strain, and tested bodies, legs, and saliva for infectious ZIKV. ZIKV was able to infect, disseminate, and be transmitted by *Aedes aegypti*. However, *Anopheles freeborni*, *Anopheles quadrimaculatus*, and *Culex tarsalis* were unable to be infected. We conclude that these species are unlikely to be involved in ZIKV transmission in North America. However, we should continue to examine the ability for other mosquito species to potentially act as ZIKV vectors in North America.

The impact of natural climate variability on the global distribution of *Aedes aegypti*: a mathematical modelling study.

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Aedes aegypti spread pathogens affecting humans, including dengue, Zika, and yellow fever viruses. Anthropogenic climate change is altering the spatial distribution of *Ae aegypti* and therefore the locations at risk of vector-borne disease. In addition to climate change, natural climate variability, resulting from internal atmospheric processes and interactions between climate system components (eg, atmosphere-land and atmosphere-ocean interactions), determines climate outcomes. However, the role of natural climate variability in modifying the effects of anthropogenic climate change on future environmental suitability for *Ae aegypti* has not been assessed fully. In this study, we aim to assess uncertainty arising from natural climate variability in projections of *Ae aegypti* suitability up to the year 2100. In this mathematical modelling study, we developed an ecological model in which *Ae aegypti* population dynamics depend on climate variables (temperature and rainfall). We used 100 projections of future climate from the Community Earth System Model, a comprehensive climate model that simulates natural climate variability as well as anthropogenic climate change, in combination with our ecological

model to generate a range of equally plausible scenarios describing the global distribution of suitable conditions for *Ae aegypti* up to 2100. Each of these scenarios corresponds to a single climate projection, allowing us to explore the difference in *Ae aegypti* suitability between the most-suitable and the least-suitable projections. Our key finding was that natural climate variability generates substantial variation in future projections of environmental suitability for *Ae aegypti*. Even for projections generated under the same Shared Socioeconomic Pathway (SSP) scenario (SSP3-7.0), in 2100 climatic conditions in London might be suitable for *Ae aegypti* for 0-5 months of the year, depending on natural climate variability. Natural climate variability affects environmental suitability for important disease vectors. Some regions could experience vector-borne disease outbreaks earlier than expected under climate change alone. Engineering and Physical Sciences Research Council and Wellcome Trust.

Secondary vectors of Zika Virus, a systematic review of laboratory vector competence studies.

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After the unprecedented Zika virus (ZIKV) outbreak in the western hemisphere from 2015-2018, *Aedes aegypti* and *Ae. albopictus* are now well established primary and secondary ZIKV vectors, respectively. Consensus about identification and importance of other secondary ZIKV vectors remain. This systematic review aims to provide a list of vector species capable of transmitting ZIKV by reviewing evidence from laboratory vector competence (VC) studies and to identify key knowledge gaps and issues within the ZIKV VC literature. A search was performed until 15th March 2022 on the Cochrane Library, Lilacs, PubMed, Web of Science, WHOLIS and Google Scholar. The search strings included three general categories: 1) "ZIKA"; 2) "vector"; 3) "competence", "transmission", "isolation", or "feeding behavior" and their combinations. Inclusion and exclusion criteria has been predefined and quality of included articles was assessed by STROBE and STROME-ID criteria. From 8,986 articles retrieved, 2,349 non-duplicates were screened by title and abstracts, 103 evaluated using the full text, and 45 included in this analysis. Main findings are 1) secondary vectors of interest include *Ae. japonicus*, *Ae. detritus*, and *Ae. vexans* at higher temperature 2) *Culex quinquefasciatus* was not found to be a competent vector of ZIKV, 3) considerable heterogeneity in VC, depending on the local mosquito strain and virus used in testing was observed. Critical issues or gaps identified included 1) inconsistent definitions of VC parameters across the literature; 2) equivalency of using different mosquito body parts to evaluate VC parameters for infection (mosquito bodies versus midguts), dissemination (heads, legs or wings versus salivary glands), and transmission (detection or virus amplification in saliva, FTA cards, transmission to neonatal mice); 3) articles that fail to use infectious virus assays to confirm the presence of live virus; 4) need for more studies using murine models with immunocompromised mice to infect mosquitoes. Recent, large collaborative multi-country projects to conduct large scale evaluations of specific mosquito species represent the most appropriate approach to establish VC of mosquito species.

Modes of Transmission of Zika Virus.

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For >60 years, Zika virus (ZIKV) has been recognized as an arthropod-borne virus with *Aedes* species mosquitoes as the primary vector. However in the past 10 years, multiple alternative routes of ZIKV

transmission have been identified. We review the available data on vector and non-vector-borne modes of transmission and interventions undertaken, to date, to reduce the risk of human infection through these routes. Although much has been learned during the outbreak in the Americas on the underlying mechanisms and pathogenesis of non-vector-borne ZIKV infections, significant gaps remain in our understanding of the relative incidence of, and risk from, these modes compared to mosquito transmission. Additional research is urgently needed on the risk, pathogenesis, and effectiveness of measures to mitigate non-vector-borne ZIKV transmission.
